Worksheet — Incline with a Pulley Simulation

Use with *Simulation - Incline with pulley.html* · Newton's 2nd law (Blocks 3–4) · goes with card C2★★ and the Block 3 Smart Cart demo

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/04-incline-pulley** in a browser (offline: *Simulation - Incline with pulley.html*). Set the surfaces, m_1, m_2 and θ, then press ▶ **Release**. The **Tasks** tab has the same three parts as this sheet; write your **explanations** here. **Part 1** before the lesson (≈ 15 min) · **Part 2** with the Block 3 Smart Cart demo (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use g = 9.81 N kg^{-1}. Take 'the hanging mass falls, the block moves up the slope' as positive.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. Card C2★★ (4.0 kg on a smooth 30° slope, 3.0 kg hanging): which way does it move?			
2. While it accelerates, is T less than, equal to or more than m_2g?			
3. Smooth slope, m_2 = 2.0 kg (m_2g = m_1g sin 30°). What happens after release?			
4. Wood slope (μs = 0.41), m_1 = 4.0 kg: which hanging masses keep it at rest?			

5. Draw separate free-body diagrams for the block and for the hanging mass of card C2★★ while they move.

Free-body diagrams (block | hanging mass)

6. Use Newton's 2nd law on each body to calculate the acceleration and the tension for card C2★★. Check with the sim.

Working

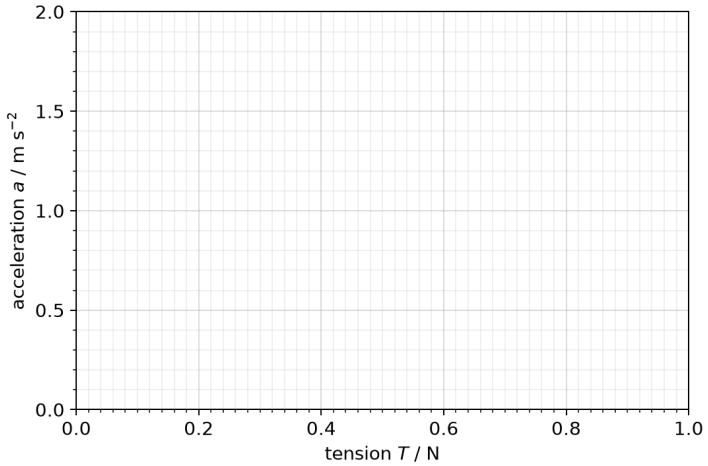
a = _____ m s^{-2} T = _____ N

Part 2 — During the lab: Block 3 Smart Cart demo

Sim: Smart Cart wheels on bench, m_1 = 0.500 kg, θ = 0°. Fill in the sim and the demo's shared data.

m_2 / g	Sim a / m s^{-2}	Sim T / N	Demo a / m s^{-2}	Demo T / N	m_2g / N
20					
40					
60					
80					
100					

7. Plot a against T (demo or sim data) and draw the best-fit line.



8. Gradient = _____ kg^{-1} ⇒ mass of the cart = _____ kg. x-intercept = _____ N = _____

9. Why is the tension while the cart accelerates smaller than m_2g? Why do we plot a against T, not against m_2g?

Part 3 — Homework: analyse

Wood on bench (μ_s = 0.41, μ_d = 0.30), m_1 = 1.0 kg, θ = 35°.

10. Find the smallest and the largest hanging mass that keep the block at rest. Draw the FBD of the block for each case — which way does static friction act?

Free-body diagrams and working

$m_{2,min}$ = _____ kg $m_{2,max}$ = _____ kg

11. m_2 = 1.5 kg. Calculate the acceleration and the tension.

Working

12. m_2 = 0.10 kg: the block slides **down**. Calculate the size of the acceleration and the tension. Why is T now **greater** than m_2g?

Working